

To Review – A+ Plan

September 8, 2026

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Built from all lecture notes and homework assignments on file as of September 8, 2026. For each course: where the course is, the textbook work that directly anchors the lectures and active homework, and the extra moves that turn understanding into an A+.

The system (all five courses)

- ☐ **Same-day proof pass:** within 24h of each lecture, re-derive every proof in the notes from scratch on paper. Anything not reproducible becomes a textbook section to re-read (listed below).
- ☐ **Problem quota:** 3–5 textbook exercises per course per week *beyond* graded homework. Log solved ones in the course `solutions.tex`, then run `./scripts/sync.sh`.
- ☐ **Homework early:** finish each HW two days before the deadline; spend the last day office-hoursing anything shaky.
- ☐ **Pre-read the next section** (listed per course) so every lecture is your second exposure, not your first.

Math 104 – Ross, Elementary Analysis (2nd ed.)

Where the course is: Lectures 01–02 (sequences, the ϵ – N definition of convergence, uniqueness of limits, the ϵ – N proof technique). HW1 was due Sep 6 (induction, irrationality, triangle-inequality equalities); HW2 due Sep 13 (sup/inf, diameter, Archimedean property + well-ordering, sequence limits, ϵ – N proofs).

Do now

- ☐ §1 The Natural Numbers – mathematical induction and well-ordering (HW1 P1, P2, P5; HW2 P2).
- ☐ §2 Some Set Notation – interval and bounded-set language for HW2 P3.
- ☐ §3 The Set $\mathbb{R}$ of Real Numbers – order, absolute value, triangle inequality (HW1 P4, P5).
- ☐ §4 The Completeness Axiom – sup/inf and the Archimedean property (HW2 P1, P2, P3).

- ☐ §7 Limits of Sequences – the ϵ - N definition; reprove uniqueness of limits yourself (Lecture 01).
- ☐ §8 A Discussion about Proofs – the ϵ - N technique is all of Lecture 02; drill Exercises 8.1–8.6 (especially 8.2: determine the limit, then prove it).
- ☐ §9 Limit Theorems for Sequences – HW2 P4 lets you use these; do Exercises 9.1–9.3.

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- ☐ Skim §5 (the symbols $+\infty$, $-\infty$; Lecture 01's limit notation). Skip §6 (optional construction of $\mathbb{R}$).
- ☐ Pre-read §10 Monotone Sequences and Cauchy Sequences – next lecture territory.
- ☐ For each ϵ - N proof, practice choosing N *before* writing scratch work, then verify – that is exam format.

Math 110 – Axler, Linear Algebra Done Right (4th ed.)

Where the course is: no lecture notes for this course – the homework packet drives the mapping. HW1 (suggested Sep 4): sets, fields, vector spaces, subspaces. HW2 (suggested Sep 11): linear independence, span, bases, dimension.

Do now

- ☐ §1A $\mathbb{R}^n$ and $\mathbb{C}^n$, including the Digression on Fields (p. 10) – HW1 P3 (no zero divisors in a field), P7 ($\mathbb{F}_2$).
- ☐ §1B Definition of Vector Space – HW1 P4 ($av = 0$ forces $a = 0$ or $v = 0$).
- ☐ §1C Subspaces (with sums and direct sums) – HW1 P5, P6.
- ☐ §2A Span and Linear Independence – HW2 P1 (LADR §2A Ex 11), P2 ($\mathbb{F}^\infty$).
- ☐ §2B Bases – HW2 P3 (LADR §2B Ex 5).
- ☐ §2C Dimension – HW2 P4 (subspaces of $\mathbb{F}^2$), P5 (LADR §2C Ex 12, $\dim(U + W)$).

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- ☐ Work every unassigned exercise in §1A–§1C and §2A–§2C: Axler's are short, and later HWs (isomorphisms, quotients, duality) build on fluency here.
- ☐ Pre-read §3A–§3B (linear maps, null spaces, ranges) – that is HW3 territory (packet p. 3).
- ☐ In $\mathbb{F}^\infty/\mathbb{F}_2$ problems, write the field explicitly every time – most lost points here are field-of-scalars mistakes.

Math 113 – Fraleigh, A First Course in Abstract Algebra (7th ed.)

Where the course is: Lecture 02 = §4 Groups (axioms, cancellation, unique solutions of $ax = b$, unique inverses, $(ab)^{-1} = b^{-1}a^{-1}$); Lecture 03 = §5 Subgroups (subgroup test, order, V_4 , GL_n/SL_n , cyclic subgroups $\langle a \rangle$). HW1 was due 9/1 (§0, §2, §3); HW2 due **today by 2PM** (§4, §5).

Do now

- ☐ §0 Sets and Relations – HW1: 0.6, 0.16, 0.30; practice 0.5–0.36.
- ☐ §2 Binary Operations – HW1: 2.2, 2.8; practice 2.3–2.37.
- ☐ §3 Isomorphic Binary Structures – HW1: 3.2, 3.30; practice 3.3–3.33.
- ☐ §4 Groups – Lecture 02 material; HW2: 4.6, 4.10, 4.18, 4.32; practice 4.1–4.33.
- ☐ §5 Subgroups – Lecture 03 material; HW2: 5.2, 5.8, 5.16, 5.50; practice 5.1–5.57.

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- ☐ Turn HW2 in *today* (due 2PM on Gradescope).
- ☐ Redo the Lecture 02/03 proofs blind: cancellation, unique solution of $ax = b$, unique inverses, subgroup test, $SL_n \leq GL_n$, $\langle a \rangle$ is a subgroup containing every subgroup that contains a .
- ☐ Pre-read §6 Cyclic Groups – Lecture 03 ended right at generation and cyclic subgroups.
- ☐ Odd-numbered exercises have solutions in the back of the book – use them to check, never to copy.

Math 118 – Boggess & Narcowich, A First Course in Wavelets with Fourier Analysis (2nd ed.)

Where the course is: Lecture 01 = complex preliminaries + the spaces ℓ^2 , L^2 (§0.3); Lecture 02 = norms, inner products, Cauchy–Schwarz (proofs now written out in the notes), orthogonality (§0.2, §0.4, §0.5); Lecture 03 = orthogonal complements *and projections* (§0.5.2), linear operators $T : \mathbb{C}^N \rightarrow V$, functionals and the dual space, convolution, Green’s functions (§0.6). HW1 due **tomorrow, Sep 9**: reverse triangle inequality for norms + B&N Chapter 0 exercises #1, 3, 9, 10, 11, 17, 23 (pp. 34–37).

Do now

- ☐ §0.2 Definition of Inner Product – HW1 #1 is literally verifying Definition 0.1.
- ☐ §0.3 The Space L^2 – 0.3.1 (definitions of ℓ^2 and L^2 ; Lecture 01’s $\ell^1 \subset \ell^2$ and L^1 vs L^2 examples) and **0.3.2 Convergence in L^2 versus uniform convergence** – the biggest nuance of the course so far.
- ☐ §0.4 Schwarz and Triangle Inequalities – read alongside the proofs section at the end of the Lecture 02 notes.

- ☐ §0.5.1 Orthogonality – orthonormal sets; HW1 #23 (orthonormal $\Rightarrow$ independent) and #17.
- ☐ §0.5.2 Orthogonal Projections – now fully in the Lecture 03 notes ($P(u) = \sum_n \langle u, e_n \rangle e_n$, best approximation, $V = V_0 \oplus V_0^\perp$); HW1 #9, #10, #11 are orthogonal-complement computations.
- ☐ §0.6.1 Linear Operators – Lecture 03’s operators, the range-of- T proposition, functionals, and the dual space.
- ☐ Finish HW1 (due tomorrow).

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- ☐ Chapter 0 exercises beyond the HW (pp. 34–38): #2, #4, #5 (verify the ℓ^2/L^2 inner products) and especially #6, #7, #8 – sequences converging in L^2 but not uniformly or pointwise. These nail the “continuous vector spaces” nuance.
- ☐ Work through the **Topics to review** list at the bottom of the Lecture 03 notes (Green’s function derivation: continuity + jump condition; verifying $u(x) = \int_0^1 G(x, y) f(y) dy$ solves the BVP). Green’s functions are professor-added material – the notes are the only source.
- ☐ Pre-read §0.6.2 Adjoints (likely next lecture) and §0.7 Least Squares (it is the payoff of orthogonal projections).

Stat 150 – Durrett, Essentials of Stochastic Processes (3rd ed.)

Where the course is: Lecture 01 = Markov property, transition matrices, random walks (reflecting, on $\mathbb{Z}$, on a circle) with a first look at recurrence/transience; Lecture 02 = classification of states (accessibility, communication classes, Chapman–Kolmogorov), first return times T_y , return probabilities ρ_{yy} , recurrent vs transient, strong Markov property, stopping times. No homework assigned yet.

Do now

- ☐ §1.1 Markov Chain Examples – map each walk from Lecture 01 onto one of Durrett’s examples (gambler’s ruin, Ehrenfest urn, weather).
- ☐ §1.2 Multi-Step Transition Probabilities – Chapman–Kolmogorov and the Markov/strong Markov properties (the “1.2 Durrett” block in the Lecture 02 notes).
- ☐ §1.3 Classification of States – ρ_{yy} , recurrent vs transient; the geometric-returns argument is now written out in the Lecture 02 notes.
- ☐ §1.4 Stationary Distributions, especially §1.4.1 Doubly Stochastic Chains – the circle-walk matrix from Lecture 01 is the canonical example.

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- ☐ Redo two computations yourself: transience of the $\mathbb{Z}$ walk when $p \neq \frac{1}{2}$ (§1.3), and why the symmetric walk still has infinite expected return time.

- ☐ A few classification exercises from §1.13.
- ☐ Pre-read §1.6 Limit Behavior – where classification starts paying off.

Date-sensitive items this week

- ☐ **Tue 9/8, 2PM:** Math 113 HW2 (Fraleigh §4–§5).
- ☐ **Wed 9/9, 11:59PM:** Math 118 HW1 (norms + B&N Ch. 0 exercises).
- ☐ **Sun 9/13, 11:59PM:** Math 104 HW2 (sup/inf, Archimedean, ϵ - N).
- ☐ **Fri 9/11 (suggested):** Math 110 HW2 (span, bases, dimension).

Prompt to regenerate this plan

CONTEXT: ~/Downloads/fa26_books holds my Fall 2026 LaTeX notes: per-course folders (math104, math110, math113, math118, stat150) with lectures/*.tex and homework/ (assignments + my solutions), plus textbooks at the repo root: math104 = Ross, Elementary Analysis 2e; math110 = Axler, Linear Algebra Done Right 4e; math113 = Fraleigh, A First Course in Abstract Algebra 7e; math118 = Boggess & Narcowich, A First Course in Wavelets with Fourier Analysis 2e; stat150 = Durrett, Essentials of Stochastic Processes 3e. Scripts: sync.sh builds changed .tex with latexmk/lualatex and commits/pushes – I run it myself; do not commit.

TASK: regenerate my per-course "To Review – A+ plan" for today's date.

1. Read every <course>/lectures/*.tex (including the newest) and every assignment PDF in <course>/homework/ (use pdftotext; assignments list exact textbook exercises and due dates). For math110 there are no lectures: the homework packet hw_packet.pdf drives the mapping; treat the two most recent HWs as active and ask me if unclear.
2. Map each lecture's topics and each homework problem to EXACT textbook sections: extract each textbook's table of contents with pdftotext and match – never guess section numbers. If a lecture topic is professor-added material not in the book (e.g., Green's functions in math118), say so and point at the lecture notes instead.
3. Filter by dates: homework past due is no longer "active" but its sections stay on the review list.
4. Write review/to_review_YYYY-MM-DD.tex (lualatex article, 11pt, enumitem checkbox lists – copy the preamble from review/to_review_2026-09-08.tex). One section per course: where the course is (lectures + active HW with due dates), a "Do now" checklist of exact sections with associated exercises (HW practice lists + recommended extras from those sections), and an "A+

- edge" checklist (extra problems, proofs to redo blind, next sections to pre-read). Include a date-sensitive deadlines section and end with this prompt verbatim in an appendix so it can be reused.
5. Build it: `latexmk -lualatex -interaction=nonstopmode -halt-on-error -cd review/to_review_YYYY-MM-DD.tex`
 6. If any lecture file still has placeholder scaffold text or notation that mismatches the textbook (like P_{yy} vs Durrett's ρ_{yy}), flag it and offer to fix it the way stat150 lectures 01-02 and math118 lecture_03 were upgraded.
 7. Do not commit or push. Reply with the PDF path and a per-course summary of what changed since the previous review document.